V.L Memorial Public School

475 Vijay Nagar

CHAPTER-8-QUESTIONS-CLASS7TH-SCIENCE

Class 07 - Science (Curiosity)

Section A

1. If a body covers equal distances in equal intervals of time, it is said to be in: **[1]**

   a) Oscillatory motion
   b) Uniform motion
   c) Non-uniform motion
   d) Random motion

2. Which of the following is a unit of speed? **[1]**

   a) s
   b) m/s
   c) kg
   d) m^2

3. Which of the following is a non-uniform motion? **[1]**

   a) A train slowing down
   b) A car moving with constant speed
   c) Earth rotating on its axis
   d) A pendulum swinging

4. If a car covers 100 km in 2 hours and 150 km in next 3 hours, its average speed is: **[1]**

   a) 62.5 km/h
   b) 50 km/h
   c) 55 km/h
   d) 60 km/h

5. Which field benefits from increasingly precise time-measuring tools? **[1]**

   a) Routine library organization
   b) Space exploration
   c) Basic handwriting practice
   d) Traditional painting lessons

6. Which of the following is not a unit of speed? **[1]**

   a) km/h
   b) m/s
   c) cm/s
   d) m^2/s

7. What does increasing clock precision allow scientists to measure? **[1]**

   a) Greater masses of physical objects
   b) Higher temperatures in laboratories
   c) Larger distances between objects
   d) Smaller intervals of time

8. Which application requires very precise time measurement in medicine? **[1]**

   a) Hospital furniture design
   b) Clinic building decoration
   c) Heartbeat monitoring
   d) Medicine packaging

9. Why is average speed commonly used for everyday motion? **[1]**

   a) Straight paths are impossible
   b) Motion requires equal distances
   c) Constant speed is uncommon
   d) Time cannot be measured

10. The S.I. unit of time is: **[1]**

a) Second
b) Minute
c) Day
d) Hour

11. Which of the following can be used to measure time? **[1]**

a) A leaking tap
b) Shadow of an object
c) A simple pendulum
d) All of these

12. What characterises uniform linear motion? **[1]**

a) Changing speed along the path
b) Unequal distances during equal times
c) Constant speed along the path
d) Repeated direction changes during motion

13. The slope of a distance-time graph indicates: **[1]**

a) Acceleration
b) Distance
c) Time
d) Speed

14. What is one-thousandth of a second called? **[1]**

a) A nanosecond
b) A centisecond
c) A microsecond
d) A millisecond

15. Which instrument measures the distance traveled by a vehicle? **[1]**

a) Speedometer
b) Barometer
c) Thermometer
d) Odometer

16. The motion of a simple pendulum is: **[1]**

a) Linear
b) Random
c) Circular
d) Oscillatory

17. Which field uses precise timing to determine race winners? **[1]**

a) Digital recording
b) Space exploration
c) Medical monitoring
d) Sports timekeeping

18. Time period of a simple pendulum depends upon **[1]**

a) Colour of the bob
b) Weight of bob
c) Both Weight of bob and Length of pendulum
d) Length of pendulum

19. Which of the following is a correct pair? **[1]**

a) Barometer - Measures humidity
b) Thermometer - Measures temperature
c) Speedometer - Measures distance
d) Odometer - Measures speed

20. Which instrument measures the speed of a vehicle? **[1]**

a) Speedometer
b) Odometer
c) Thermometer
d) Barometer

21. **Assertion (A):** A sundial cannot measure time at night. **[1]**
Reason (R): A sundial works based on the shadow cast by the Sun.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** The speed of a car is measured in km/h. **[1]**
Reason (R): Speed is the distance covered per unit time.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** Digital clocks are more accurate than mechanical clocks. **[1]**
Reason (R): Digital clocks use electronic oscillations, which are more precise.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** An hourglass is not used for precise time measurement. **[1]**
Reason (R): The flow of sand can vary due to external factors.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. **Assertion (A):** The distance moved by an object in unit time is termed as its speed. **[1]**
Reason (R): Faster vehicles have higher speeds.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. **Assertion (A):** Speedometer records the speed of the vehicle in km/h. **[1]**
Reason (R): Odometer measures the distance covered by the vehicle in first hour.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

27. **Assertion (A):** The basic unit of time is second(s). **[1]**
Reason (R): Quartz clock gives more accurate measurement of time as compared to other clocks.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

28. A car speeds up, then maintains speed. What type of motion does it have? **[1]**
29. What did Galileo conclude about the oscillation of a pendulum of a given length? **[1]**
30. Name one ancient device used to measure time. **[1]**
31. What does a straight line on a distance-time graph indicate? **[1]**
32. If we say that a train is moving at 40 kilometres per hour, what do we mean by this statement? **[1]**
33. What is the time period of a pendulum? **[1]**
34. Name the physical quantity which helps to know which object is faster or slower. **[1]**

Section B

35. Match the following: **[2]**

Column A	Column B
(a) Uniform motion	(i) Sloped curved graph line
(b) Distance–time graph	(ii) Shows how far object moves
(c) Non-uniform motion	(iii) Plotted on X-axis
(d) Time	(iv) Straight line on graph

36. **Match the following:** **[2]**

Column A	Column B
(a) Walking person	(i) Moves very slowly
(b) Car on highway	(ii) Moves at moderate speed
(c) Airplane	(iii) Moves very fast
(d) Tortoise	(iv) Moves faster than walking but slower than a car

37. Why do we use digital clocks today instead of sundials? **[2]**
38. Describe the structure and motion of a simple pendulum. **[2]**
39. Distinguish between uniform and non-uniform motion using the example of a car moving on a straight highway with no traffic and a car moving in city traffic. **[2]**
40. How can we make a simple pendulum experiment to measure time? **[2]**
41. The fastest galloping horse can reach the speed of approximately 18 m/s. How does this compare to the speed of a train moving at 72 km/h? **[2]**
42. A car covers 60 km in the first hour, 70 km in the second hour, and 50 km in the third hour. Is the motion uniform? Justify your Answer. Find the average speed of the car. **[2]**
43. Why is a distance-time graph useful? **[2]**

Section C

44. Explain why an hourglass is not suitable for measuring exact seconds. **[3]**
45. How does a speedometer work in vehicles? Why is it important? **[3]**
46. Explain how developments in mechanical timekeeping progressed from the fourteenth to the seventeenth century. **[3]**
47. Describe how distance-time graphs help in distinguishing between uniform and non-uniform motion. **[3]**

48. Discuss the limitations of sundials in measuring time. **[3]**

49. Explain how precise time measurement supports different activities in modern society. **[3]**

50. A vehicle moves along a straight line and covers a distance of 2 km. In the first 500 m, it moves with a speed of 10 m/s and in the next 500 m, it moves with a speed of 5 m/s. With what speed should it move the remaining distance so that the journey is complete in 200 s? What is the average speed of the vehicle for the entire journey? **[3]**

Section D

51. **Read the following text carefully and answer the questions that follow:** **[4]**

By comparing the distances moved by two or more objects in a unit time, it can be found out which of them is moving faster. The unit time may be one second or one minute or one hour. We call the distance covered by an object in a unit time as the speed of the object. It can be calculated, if we know the total distance covered by an object and the time taken to cover it. The speed of an object is the total distance covered divided by the total time taken to cover it. Thus,

$$\text{Speed} = \frac{\text{Total distance covered}}{\text{Total time taken}}$$

We know the SI units of length and time. Since the speed is distance/time, the SI unit of speed is metre/second and is expressed as m/s. Speed can also be expressed in other units. If we express the distance in kilometre and time in hour, then the unit of speed is kilometre/hour, expressed as km/h. An object moving along a straight line with a constant speed is said to be in uniform linear motion. So, the train is in uniform motion between B and C. On the other hand, if the speed of an object moving along a straight line keeps changing, it is said to be in non-uniform linear motion. The motion of the train between A and B, as well as between C and D, is non-uniform. An object in uniform linear motion covers equal distances in equal intervals of time, while it covers unequal distances in equal intervals of times when it is in non-uniform linear motion.

Questions:

a. What is the SI unit of speed? (1)

b. What is meant by uniform linear motion? (1)

c. Define speed. How can it be calculated? (2)

OR

Differentiate between uniform and non-uniform linear motion with examples. (2)

52. **Read the following text carefully and answer the questions that follow:** **[4]**

A simple pendulum consists of a small metallic ball (called the bob of the pendulum) suspended from a rigid support by a long thread. The pendulum at rest is in its mean position. When the bob of the pendulum is moved slightly to one side and released, it starts oscillatory motion. Its motion is periodic in nature because it repeats its path after a fixed interval of time. The pendulum is said to have completed one oscillation when its bob, starting from its mean position O, moves to extreme position A, changes direction and moves to another extreme position B, changes direction and comes back

to O. The pendulum also completes one oscillation when its bob moves from one extreme position A to another extreme position B and comes back to A. The time taken by the pendulum to complete one oscillation is called its time period.

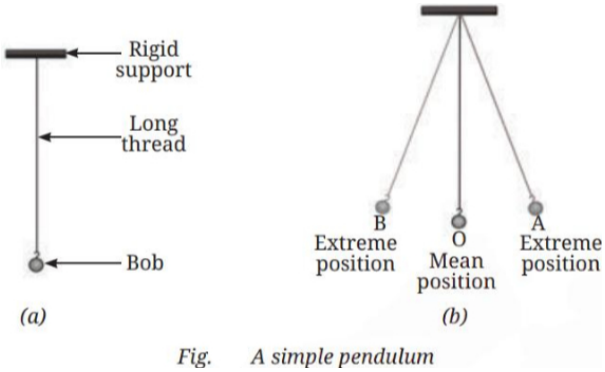


Fig. A simple pendulum

Questions

a. What is the bob of a simple pendulum? (1)

b. Why is the motion of a simple pendulum called periodic? (1)

c. What is meant by one oscillation of a simple pendulum? (2)

OR

Define the time period of a pendulum. What does it represent? (2)

53. **Read the following text carefully and answer the questions that follow:** **[4]**

Humans got interested in keeping track of time long ago. They started noticing that many events in nature repeat themselves after definite intervals of time. For example, the rising and setting of the Sun, the phases of the Moon and the changing seasons. They started using the cycles of these events for timekeeping. First, they devised calendars. A day was defined by the cycle of rising and setting of the Sun. Then began the quest to find ways of knowing the time of day.

They made many devices which helped them to measure smaller intervals of time within a day. Some of these were sundials, water clocks, hourglasses, and candle clocks. In a sundial, time is determined with the changing position of the shadow of an object cast by the light of the Sun during the day.

Questions:

a. How did humans first start measuring time using natural events? (1)

b. What was the primary basis for defining a day in early timekeeping methods? (1)

c. How does a sundial help in measuring time? (2)

OR

What were some of the devices used by ancient civilizations to measure time? (2)

Section E

54. Why is average speed sometimes misleading when describing motion? Give a scenario. **[5]**

55. Explain how in ancient time a day, a month and a year were measured? **[5]**

56. Evaluate the development of water-based timekeeping in ancient India, including the limitation that led to the sinking bowl water clock. **[5]**

57. How has the measurement of time evolved from ancient to modern methods? Discuss the advantages of modern timekeeping devices. **[5]**

58. Discuss why the atomic definition of a second is important for modern science and technology. **[5]**

59. How can understanding relative speeds help in planning everyday travel? Provide an example. **[5]**

Section F

60. **State whether the given statement is True or False:** **[8]**

(a) Speed is calculated by dividing distance by time. **[1]**

(b) If a distance–time graph shows a curved line, the motion is uniform. **[1]**

(c) A day is slightly longer than 24 hours due to Earth's rotation. **[1]**

(d) Uniform motion can still occur if the object is accelerating gradually. **[1]**

(e) Uniform motion means an object covers equal distances in equal intervals of time. **[1]**

(f) A minute consists of 100 seconds. **[1]**

(g) The SI unit of time is the second. **[1]**

(h) If you travel 150 km in 3 hours, your average speed is 50 km/h. **[1]**

61. **Fill in the blanks:** **[8]**

(a) The time taken to complete one oscillation is called the _____. **[1]**

(b) A train moving at 80 km/h throughout a journey exhibits _____ motion. **[1]**

(c) A day is divided into _____ hours. **[1]**

(d) The motion of Earth around the Sun is an example of _____ motion. **[1]**

(e) A cheetah is _____ than a human when comparing usual running speeds. **[1]**

(f) The distance-time graph of an object at rest is a _____ line. **[1]**

(g) A device used to measure time is called a _____. **[1]**

(h) The distance covered by an object in _____ is called its speed. **[1]**